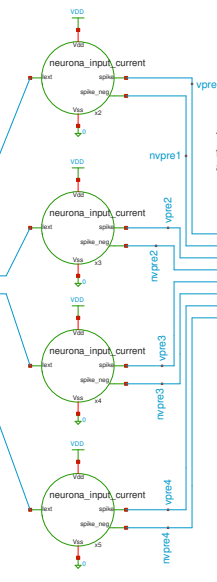


An input signal for testing purposes and its complement signal, to be processed by the encoder, providing excitation current for the input neurons



4 presynaptic neurons



```

s1
.tran ln 200u
.save all
.control
run
write tb_full_4x2.raw
.endc

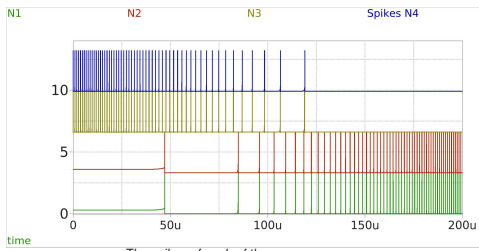
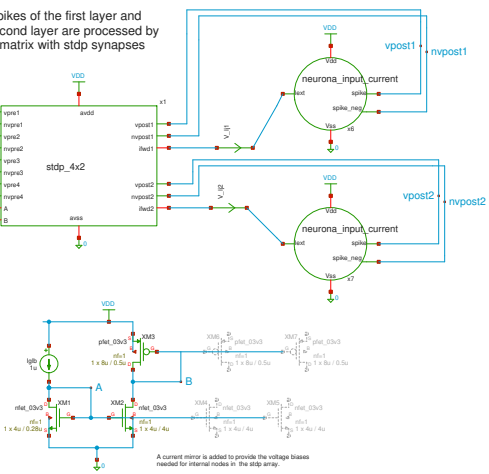
MODELS
.include $:/180MCU_MODELS/design.ngspice
.lib $:/180MCU_MODELS/sml41064.ngspice typical
.lib $:/180MCU_MODELS/sml41064.ngspice cap_mis
.lib $:/180MCU_MODELS/sml41064.ngspice res_typical
.lib $:/180MCU_MODELS/sml41064.ngspice moscap_typical
.lib $:/180MCU_MODELS/sml41064.ngspice mimcap_typical
* .lib $:/180MCU_MODELS/sml41064.ngspice res_statistical

```

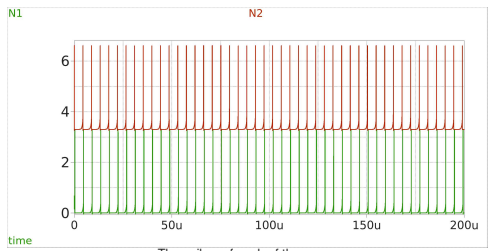
load waves

The spikes of the first layer and the second layer are processed by a 4x2 matrix with stdp synapses

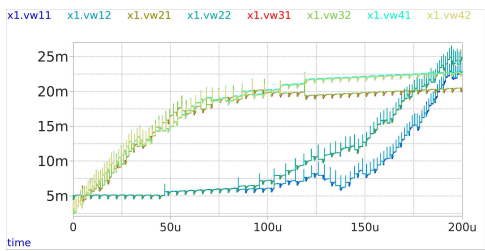
two post synaptic neurons



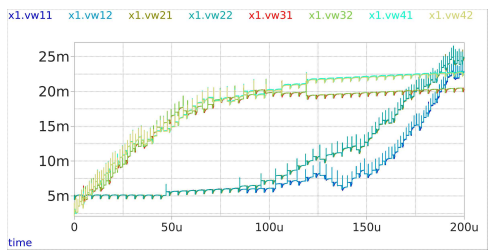
The spikes of each of the neurons at the first layer. The encoder is set to use population encoding, which is that not all the neurons respond equally to the same input stimuli



The spikes of each of the neurons at the output layer. The synapses provide enough input current for each of the postsynaptic neurons



The synaptic weights connected to the first postsynaptic neuron.



The synaptic weights connected to the second postsynaptic neuron.